Projective Derivation Calculus: Checked Readouts and a Failed Truth Test

Working draft

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Abstract

Projective Derivation Calculus (PDC) extends the decode stratum of Projective Incidence Calculus (PIC) with position-indexed reads, alternative bodies of an atom, explicit Viterbi and marginal readouts, and an external theory judgment. Its one-position, one-body vocabulary fragment is kernel-checked against PIC. Finite artifacts retain the body table, readout, margin, fragility, provenance and covered candidate set. Frozen GPT-2 measurements identify the read mixture on 108 head/query instances and exhibit readout fragility on competing copy routes. A fixed-computation admission oracle admits a positive control, refuses a rational three-context control, and finds no singleton irreducibility certificates on 80 Qwen positions. An independently checked addition experiment then triggers the program's preregistered epistemic stop rule: among certified commitments, 7/8 Viterbi answers and 9/10 marginal answers are false in the finite theory. The margin certificate preserves a covered score decision; that property does not establish truth on this domain. Admission learning is negative, and the final held-out service audit has zero coverage. Admitted-memory transfer and useful service remain unestablished. This paper records the stopped stage and does not claim the full program is complete.

1 Objects and the PIC seam

[definition] A source contributes an incidence to an atom. A body is a finite source coalition; its bracket adds those incidences. For a biased decoder, the executable complete-body bracket includes the atom bias once. Thus a body's annotation is temperature independent. Alternatives of an atom are combined by the temperature family

$$\alpha_T(a) = T \log \sum_{P \in \mathrm{deriv}(a)} \exp\big([P](a)/T\big), \qquad T > 0.$$

The two readouts are

$$\mathrm{Vit}(a) = \max_{P \in \mathrm{deriv}(a)} [P](a), \qquad \mathrm{Mar}_T(a) = \alpha_T(a).$$

An empty body set has negative-infinity score and cannot win. Winner sets retain ties; a tied readout does not commit an atom. Body identifiers retain multiplicity when distinct bodies have equal brackets. These sums can count shared subbodies repeatedly. No possible-world or disjoint-explanation probability semantics is asserted.

[proved] Restrict the chart to one position, vocabulary atoms, at most one body per atom, and no external theory. For the singleton full-source coalition the annotation is PIC's biased logit

$$\alpha_T(v) = [S](v) = L(v) = \left\langle \sum_{j \in S} d_j, U_v \right\rangle + b_v.$$

In the Isabelle representation, a present score is `Some` of this real value; `None` represents an absent atom. The equality to a finite logit is the displayed singleton case. Empty atoms are not assigned a finite logit by this translation. Positive-temperature softmax preserves logit order and argmax sets. Empty/singleton body tables have equal Viterbi and marginal annotations, so their winner sets are temperature independent.

These statements import the actual PIC definitions, rather than a parallel logit semantics. Their artifacts are `i-orca/examples/pdc/PDC_Derive.thy`: `mar_singleton`, `one_body_agreement`, `one_body_temperature_invariance`, and the four `pic.specialization_alpha_eq_logit`, `pic.specialization_alpha_eq_residual`, `pic.specialization_softmax_order`, and `pic.specialization_softmax_argmax` lemmas. The strict session is `PDC`, parent `PIC_Core`; `quick_and_dirty=false`, zero proof holes.

[witness, proved] Let atom A have two distinct bodies of score zero and atom B have one body of score $1/2$. At $T = 1$,

$$\mathrm{Vit}(A) = 0 < 1/2 = \mathrm{Vit}(B), \qquad \mathrm{Mar}_1(A) = \log 2 > 1/2 = \mathrm{Mar}_1(B).$$

The kernel lemmas `witness_viterbi`, `witness_marginal` and `witness_fragile` check both winners and their disagreement. The one-body inheritance stops when alternatives are summed inside an atom.

2 Reads, bodies and finite exports

[definition] A head reads key positions through incidences $i \triangleright_h j = \langle q_i, k_j \rangle$. Under a scaled-query convention the native GPT-2 score is this incidence divided by $\sqrt{d_h}$. The head write is its softmax-weighted value mixture. Its output projection is a further linear write; the shared projection bias belongs to the block. Value mixtures, vocabulary decisions and external truth are distinct objects in the artifact.

[empirical, R2] Before capture we fixed three token-ID contexts, layers $0, 5, 11$, heads $0, 3, 7, 11$, and query positions $3, 7, 15$. The 108 cases use the pinned f16 GPT-2 bundle with native f32 arithmetic. The observer reads the arrays assigned by the actual forward. The native finite causal mask is -10^{10}; forbidden-key probability underflows to zero on these instances. This numerical convention is recorded explicitly. The assignment itself is the masked-softmax/value mixture; f64 recomputation and key agreement are separately measured:

R2 quantity	Result
Scaled-incidence maximum absolute error	$7.143 \cdot 10^{-6}$
Probability maximum absolute error	$2.983 \cdot 10^{-7}$
Head-write maximum absolute / RMS error	$6.683 \cdot 10^{-7}$ / $4.239 \cdot 10^{-8}$
Incidence/native-probability argmax-set agreement	108/108
Independent NumPy vocabulary top-1 agreement	48/48
Observer hidden-state maximum error	0

All 12 named heads pass on this set. No other architecture or prompt distribution inherits that identification. See `fieldrun/experiments/pdc/R2_PREREG.md` and `R2_RESULTS.md`.

[empirical, R3] A second preregistration fixes 16 contexts with competing continuations of a repeated trigram. Every earlier exact match and each of the 12 heads generates a body: the actual weighted value write at that continuation, projected through its native output matrix and read in the measured final-LayerNorm-folded frame. No route is selected using the host token.

There are 576 bodies, 36 per context. Viterbi agrees with the full host on 8/16 contexts and marginal on 6/16; the readouts disagree on 2/16. Both agree with the trigram task label on 10/16. Independent NumPy predictions agree with all 256 native per-position vocabulary predictions. These are candidate copy routes, with no causal sufficiency or necessity claim. See `R3_PREREG.md`, `R3_RESULTS.md`, and the complete `r3_charts.json` body/source artifact.

[empirical, R1] The finite executor retains readout names, full body tables and ties. It reproduces the checked witness and matches the same PIC argmax on 256 generated one-body programs at five fixed temperatures (1,280 evaluations). Its float implementation is not itself a kernel certificate; absent, tied and nonfinite inputs have explicit behavior. See `fieldrun/scripts/pdc_derive.py` and `R1_RESULTS.md`.

3 Admission and its limits

[definition] Crystallization admits an atom as a source only under a margin rule. A common score offset does not affect a decision. If the coalition margin exceeds $\gamma > 2\delta$ and the admitted source's score errors, after a common offset, are bounded by δ at the target and every covered rival, its decision survives. The omitted-coalition form applies the same premise to the tail incidences. The candidate set must remain unchanged throughout the certificate.

[proved] `i-orca/examples/pdc/PDC_Admission.thy` packages PIC's pointwise margin certificate as `admitted_source_preserves_decision`, `pic.crystallization_admitted_write`, and `pic.crystallization_complement_certified`. A fourth lemma, `pairwise_source_preserves_decision`, permits a guard per rival. For a target t and rival v, the body gap must exceed the absolute drift of that pair's gap from coalition scores to the source scores. Each pair can use its own offset and δ; no whole-vocabulary perturbation ball is introduced. All four lemmas pass the strict Isabelle session. Their conclusions are score decisions, not truth judgments.

Scope of all kernel claims. The kernel results concern finite score identities, readout comparisons and preservation of covered score decisions under explicit premises. They do not establish external-theory truth, useful reasoning, or agreement with an unrestricted full-vocabulary host outside the covered candidate set. A nonzero admission rate alone would establish none of those stronger claims. Full-host agreement and external-theory judgments require their own independently checked evidence; protected-bank preservation is not generalization. R7's fixed-budget negative does not identify a unique bottleneck or prove that admission is unlearnable.

[empirical, R4] The oracle fixes measured source writes and frees one shared affine decoder frame. Faithfulness and a tested singleton's decision are linear inequalities. Feasible frames and solver statuses are retained; infeasibility requires checked Farkas multipliers. Solver errors remain unknown. Admission is checked separately from LP reducibility.

Domain	Reducible	Admitted	Irreducible
Positive synthetic corpus, N=64	64	64	0
Three-context refusal control, hard context	0	0	1
Qwen held-out regime rerun, N=80	80	11	0

The positive control stores its feasible native frame. The refusal control has exact rational singleton Farkas checks. All 80 Qwen physical frame witnesses are checked independently of the solver. Maximum faithfulness violation is $1.844 \cdot 10^{-10}$, within the fixed 10^{-7} numerical tolerance. Eleven first-feasible frames pass the stricter centered-tail admission guard; 69 do not. The latter are not universal impossibility certificates over all frames or coalitions.

The program's historical "0/40" premise required correction: it meant zero irreducibility certificates, with every position reducible. The new sample likewise has an empty irreducibility-certificate set at N=80. The missing historical dump is not claimed to have been replicated bitwise. The expectation that this is an established negative admission domain is retired. See `pil/docs/notes/pdc_r4_prereg.md`, `pdc_r4_outcome.md`, and `pil/results/pdc_r4/verification.json`.

4 External truth: a preregistered failure

[definition] $P \vdash a$ records a source-coalition score decision. $\mathcal{T} \models a$ is a separate decidable judgment. A recorded commitment can be affirmed by both, or be a hallucination relative to the theory. A miss occurs when the theory affirms an atom that no recorded body strictly decides. A fragile truth is selected marginally but not by Viterbi. None of these judgments is recovered merely by renaming a host margin.

[empirical, R5] The fixed theory is ordinary integer addition for operands 0–4 and answers 0–8. Three fixed prompt templates cover all 25 ordered pairs, giving 75 evaluation contexts. A fourth template's 25 contexts were reserved until the final audit below. Leading-space answer tokens are fixed before any forward call. Each atom has three fixed bodies: the full source coalition, the attention-block coalition, and the MLP-block coalition. The 2,025 annotations are retained in the exported bundle.

The advance threshold is $\gamma = 0.5$. Both the named readout and its chosen best-bracket body's margin must clear this threshold. Each covered rival must also pass the pairwise drift guard. Ties, coverage mismatches and failed guards abstain. The theory does not enter that selection. The hallucination bar is strictly below 10%; zero commitments is untested. The preregistered wide-miss threshold is 25%.

R5 outcome	Viterbi	Marginal
Certified commitments	8	10
Abstentions	67	65
Theory-affirmed commitments	1	1
Hallucinations	7	9
Hallucination rate among commitments	87.5%	90%
Raw readout/theory agreement	16%	20%
Committed/full-host agreement	62.5%	60%
Committed/covered-candidate host agreement	100%	100%

There are 47 misses, three fragile contexts, and three fragile truths. Both readouts miss the advance bar widely. Both finite-domain epistemic claims are retired; the R5 stop restriction is active.

[empirical] Native and independent NumPy final-position predictions agree 75/75; the observer's hidden-state effect is zero. Raw source-sum, final-LN-folding and full-body candidate-score maximum errors are, respectively, $1.682 \cdot 10^{-5}$, $1.702 \cdot 10^{-4}$ and $2.399 \cdot 10^{-4}$. The finite pairwise check uses actual native candidate scores, so those discrepancies are not hidden behind an assumed global epsilon. A separate auditor reconstructs every source/body annotation, both readouts and every commit/abstain decision. Maximum annotation discrepancy is zero. A separate theory checker computes addition from the operands and rejects a purported commitment lacking a body decision.

One retained case makes the limitation concrete: `t2-a0-b0` asks for $0 + 0$ under the ordinary-addition prompt. Both readouts commit 1 using the MLP coalition; the unrestricted native host also predicts 1. The body margin is 3.040224 and minimum pairwise guard slack is 1.040681. The

certificate holds while the theory rejects the answer. An attention body in the same chart decides 0, so this case is not a miss. The negative result concerns this frozen model, domain, prompts, candidate set, bodies and threshold. It is not an impossibility theorem about all models.

The full evidence is `fieldrun/experiments/pdc/R5_PREREG.md`, `R5_RESULTS.md`, `r5_bundle.json`, `r5_audit.json`, and `r5_theory_results.json`. Their input hashes are retained. The certificate covers nine tokens; it does not certify the unrestricted vocabulary argmax, whose agreement is separately reported above.

5 The stopped program and remaining obligations

[empirical, R6] A fixed four-arm comparison now measures all 27 triples in $\{0,1,2\}^3$ for $(a+b)+c$. It reuses nine first-hop R5 captures and freezes exact token assembly, memory vectors and admission attempts before any new hop-2 output. All nine first hops are refused by the recorded native-frame singleton procedure, giving 27 refused triples and no admitted stratum. Re-derive and crystallize each score 3/27; generated append and tool append each score 9/27. Generated append retains all three baseline-correct cases and adds six, a 22.22-point gain above the advance five-point bar. This supports the operational comparison on this refused slice. Admitted redundancy and actual admitted replay remain untested. The tool and generated arms have equal aggregate accuracy but different correctness on four cases; no pointwise tool upper bound or semantic-content necessity is proved.

Independent NumPy final-position predictions agree 108/108. Every refused crystallize case is exactly identical to re-derive and injects no source. A separate zero/nonzero replay engineering check finds zero-write identity, a nonzero hidden effect of 0.537842 and agreement of the induced change with NumPy within $6.390 \cdot 10^{-5}$. The baseline's absolute hidden mismatch of 0.003174 is separately retained; it is not claimed below the initially attempted 10^{-3} absolute diagnostic. These engineering probes are not admitted memory observations. See `fieldrun/experiments/pdc/R6_PREREG.md`, `R6_RESULTS.md`, and `r6_results.json`. The exported two-hop graph has 117 independently checked affine decode nodes and 108 provenance paths, with 27 generated, 27 external-theory tool and 54 no-memory edges. Its finite margin coverage is distinct from its observed unrestricted native token.

[empirical, R7 negative] Two preregistered 200-proposal arms learn an untied output frame and bias with fixed native writes and gates. Both decode-only and admission-objective arms retain zero admissions out of 75; the improvement and control-advantage gates fail. All 183 protected full-host predictions remain unchanged. All accepted steps have nonzero motion, satisfy logged per-step drift premises, and retain FP $\geq W$. Independent NumPy endpoint reconstruction agrees within $5.41 \cdot 10^{-13}$. These are protected-bank observations, not a generalization result; 25 reserved contexts remained untouched until the final audit below. See `pil/docs/notes/pdc_r7_prereg.md`, `pdc_r7_outcome.md`, and `pil/results/pdc_r7/verification.json`. R6's admitted-side claim remains unmeasured, R5 remains retired, and no chart-depth versus layer-depth function is claimed.

[empirical, saved-frame diagnostic] A separately frozen offline four-arm analysis compares original singleton admission, a halved absolute margin floor, highest-source plus bias, and random-source plus bias. All four admit zero of 75 at initialization and both R7 endpoints; all 183 protected outputs remain unchanged. At each endpoint, nine contexts have a strict singleton decision matching the covered host target without the stronger certificate, but no singleton passes $m > 2\delta$. Thus the absolute floor alone cannot explain all refusals. This identifies no unique causal learning bottleneck and examines no larger coalition. Conditional host agreement among admissions is undefined. No model call or final held-out input is used. See `pil/docs/notes/pdc_r7_diagnostic_prereg.md`,

pdc_r7_diagnostic_outcome.md, and pil/results/pdc_r7_diagnostic/verification.json. The original negative R7 gate and v0.10 archive remain unchanged.

[empirical, R8-H] A preregistered host-only continuation now runs a fresh native subset forward and returns a served bundle or an explicit abstention. A tokenizer-text rule freezes 1,703 probe tokens before their scores are inspected. Its certificate checks the chosen body's target against every probe, while the derive query retains the nine answer atoms. The independent audit checks 383,175 body/probe annotations and finds zero source-write and probe-score discrepancy against a full baseline. Native output-row dot calls fall from 3,769,275 to 127,725 across the 75 contexts, with 383,175 additional body/probe dots in each arm. No layer or position is skipped. Both readouts abstain on all 75 contexts. The required nonempty coverage gate therefore fails; host agreement and hallucination rates among commitments are undefined. This compute reduction produces no usable answers under the fixed policy. See fieldrun/experiments/pdc/R8H_PREREG.md, its timing addendum, R8H_RESULTS.md, and r8h_results.json.

One fixed ordered timing pair gives 13.355 seconds end to end for the full baseline and 12.871 for subset service; these descriptive timings do not establish a general speedup. The runtime also serves a single named non-held-out request and rejects a reserved request before execution. The final audit below subsequently consumed the 25 reserved contexts. R8's original useful truth/host service gate is [open]; theory soundness is not advertised from the failed margin instrument.

[open] This working paper records R0–R8 outcomes after the stop, including negative R7 and the final held-out diagnostic. It provides a stopped-stage evidence map in pdc/spec/EVIDENCE.md, not a passing R0–R8 program or criterion-9 result. The complete north star remains unachieved. The Eiffel-tower illustration in the specification remains an illustration and supplies no experimental evidence.

6 Artifact discipline and reproducibility

The canonical definitions and claim tags are in pdc/spec/PDC_SPEC.md. PDC contains the specification and this paper; proof sources remain in i-orca, frozen-model exports in fieldrun, and admission-oracle work in pil. The repos exchange named JSON/NPZ artifacts, without local-path code imports. [empirical, final held-out diagnostic] After fixing the R6/R7 outcomes, a one-shot preregistered audit opens the 25 reserved template-3 contexts with the unchanged original R8-H policy. Both readouts abstain on all 25. Host agreement and hallucination among commitments are undefined; the nonempty-coverage gate fails. The independent checker verifies 127,725 body/probe annotations with zero discrepancy. Native output-row calls fall from 1,256,425 to 42,575, with 127,725 body/probe dots per arm and no layer or position skipped. This is compute saving at zero useful coverage. Each bundle records memory provenance as none and external theory judgment separately. No learning endpoint was selected after its failed advance gate. See fieldrun/experiments/pdc/FINAL_AUDIT_PREREG.md, FINAL_AUDIT_RESULTS.md, and final_audit/results.json. The reserved slice is now consumed and must not be used for any future threshold, body-generator or checkpoint selection. A new confirmatory experiment requires fresh held-out data. R8 and criterion 9 remain unmet; this paper documents a stopped stage rather than achievement of the north star.

[empirical, stronger-model follow-up] A separately frozen, user-authorized Qwen3.5-4B comparison uses 100 new chat inputs, split by instruction wording into 50 development and 50 heldout cases. Both splits reuse the same 25 arithmetic facts; this is not an unseen-fact test. The host is correct on 100/100, and both readouts commit correctly on all 50 heldout cases with no abstention or hallucination. Every winning witness is the full source program; neither the attention

nor MLP body wins. Thus this is nonzero full-program coverage, not a compact sufficient circuit or a learned-admission result. Since the uncertified host is also always correct, no hallucination reduction over that baseline is measured. No native output rows, positions or layers are skipped.

The pinned checkpoint is `Qwen/Qwen3.5-4B` revision `851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a`, with float16 weights, CPU inference and thinking disabled. The new model, architecture, tokenizer and chat format prevent attribution to parameter count alone. Captured embedding plus attention/MLP writes give 65 sources; RMS folding uses the observed full-host scale with $1 + w$, without mean subtraction or bias. DeltaNet is not identified with ordinary softmax read-PIC. The fixed query has nine digit atoms; certificate coverage is 2261 tokenizer-text probes, while the native host scores all 248320 output rows. The original three bodies, $T = 1$, $\gamma = 0.5$ and exact pairwise guard are retained.

Independent Transformers verification matches all 100 host winners. The independent auditor reconstructs 678300 body/probe annotations and 200 readout commitments, with maximum discrepancy $1.60 \cdot 10^{-14}$. Native scientific and independent verification budgets are 100 forwards each. Original GPT-2 negatives, archive and the consumed final-25 ban are unchanged. See `fieldrun/experiments/pdc_qwen35_4b/PREREG.md`, `RESULTS.md`, `results.json`, and `verification.json`. This comparison meets its local gates; it does not close R8, two-hop transfer or the full north star.

[empirical, singleton diagnostic] A separately frozen saved-data analysis scores all 65 native Qwen sources on all 100 existing contexts. Every context has a singleton selecting the host target on the 2261 probes; in 97 a singleton also exceeds the absolute margin floor 0.5. Nevertheless, exact pairwise and centered-tail admission are both 0/100. An independent torch/alternative-algebra checker confirms all 6500 source verdicts within $5.69 \cdot 10^{-14}$. No new model call, larger coalition, threshold sweep, learning or replay occurs. This is not fresh held-out confirmation. See `fieldrun/experiments/pdc_qwen35_singletons/PREREG.md`, `RESULTS.md`, `results.json`, and `verification.json`.

[proved] For a host gap h and body gap p, the guard $|h - p| < p$ is equivalent to $0 < h < 2p$, as checked by `PDC_Admission.pairwise_gap_guard_iff`. Thus agreement of positive gaps alone is weaker: the body must supply more than half the host gap against every covered rival. This identifies a sufficient guard's relative scale requirement; it proves neither necessity for matching decisions nor causal sufficiency of an observed source. The original full-program positive and fixed-singleton negative are distinct results; no impossibility or certificate-to-truth claim follows.

[empirical, Qwen two-hop refused slice] A preregistered 27-triple comparison reuses nine first-hop captures and evaluates 54 unique second-hop inputs for 108 named paths. Full-host arithmetic accuracy is 12/27 for rederive and refusal-fallback crystallize, versus 27/27 for generated and tool append. Independent Transformers agreement is 54/54; an independent body audit checks 366282 annotations within $2.14 \cdot 10^{-14}$. All first hops are refused: admitted replay and admitted-stratum redundancy remain untested. Appending also changes an empty assistant turn to a nonempty one; no neutral-token or wrong-token control isolates semantic memory use. Both baseline readouts commit 25 answers, including 14 hallucinations (56%). Covered score preservation therefore still does not imply truth. See `fieldrun/experiments/pdc_qwen35_memory/PREREG.md`, `RESULTS.md`, `results.json`, and `verification.json`.

[empirical, Qwen guarded learning negative] A subsequent frozen protocol trains only 2261 output rows and biases, retaining native source writes, gates, input embeddings and backbone. Both cross-entropy control and admission-objective arms finish 200 proposals with admissions unchanged at 0/100. Independent endpoint reconstruction verifies all 208 protected native decisions, all singleton guards, and full-vocabulary $FP \geq W$; intermediate checks inspect runtime logs, not reconstructed intermediate frames. Protected decisions include 30 wrong second-hop outputs. Min-

imum protected margins shrink from 0.394895 to $1.32 \cdot 10^{-5}$ (control) and $1.89 \cdot 10^{-5}$ (admission); later rejected proposals restore prior frames. This does not isolate the preservation constraint as the sole bottleneck. The unchanged fieldrun chart-quality gate passes, but the joint advance gate fails because admission never rises. No budget extension, fresh confirmation or actual admitted replay is claimed. See `pil/docs/notes/pdc_qwen35_learning_prereg.md`, `pdc_qwen35_learning_outcome.md`, `pil/results/pdc_qwen35_learning/verification.json`, and `fieldrun/experiments/pdc_qwen35_memory/learning_chart_audit.json`.

[empirical, memory-content controls] A separately preregistered development diagnostic fixes two new one-token controls on the same 27 known triples. A period gives 9/27 correct final answers. A wrong digit $w = (a + b + 1) \bmod 5$ gives 0/27 correct, with all 27 answers equal to $w + c$. The reused empty/correct-digit baselines remain 12/27 and 27/27. Native and independent reference agree 54/54; an independent input/count auditor checks all contexts and 23 frozen input/code/model/runtime hashes. The preregistered content-sensitivity bar passes, and neutral improvement does not. Correct and wrong digits use identical answer-slot lengths and surrounding IDs: supplied-digit content affects this slice's behavior. The period is an imperfect distribution/semantics control. No exclusive internal mechanism, fresh confirmation, admitted residual replay or certificate-to-truth claim follows. See `fieldrun/experiments/pdc_qwen35_memory_controls/PREREG.md`, `RESULTS.md`, `results.json`, and `verification.json`.

Every [proved] statement above names its checked theory and lemma; every [empirical] claim names a preregistration and run. Concrete floating-point data checks are distinguished from symbolic kernel theorems.

The proofs build with Isabelle2025-2:

```
isabelle build -o timeout=40 -d i-orca/examples/pic_core -D i-orca/examples/pdc
```

Native capture commands, independent NumPy checks, the solver-free R4 verifier, and the R5 bundle/theory auditors are listed in the corresponding result documents. The finite executor and certificate/theory tests pass. The paper and companion changes are published as review PRs with versioned evidence releases; merging remains pending. R8-H is a measured negative coverage arm; no unperformed or failed positive gate is counted as completed by this paper.